Edinburgh Napier UNIVERSITY

Visualisation Components

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SET09123 – Interactive Data Visualisation

Lecture Summary

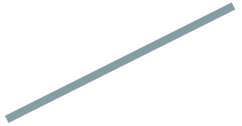
- Marks and Channels
- Gestalt Principles and Pre-attentive Features
- Coordinate Systems and Scales

Marks and Channels

Marks

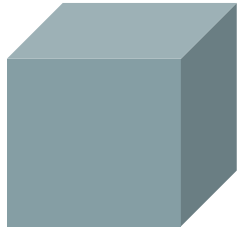
- *Marks* are the basic graphical units to represent data items (or links)

Point – 0D



Line – 1D

Polygon – 2D



Polyhedron – 3D

Channels

- *Channels* are the visual attributes used to encode data on marks

- They are sometimes labelled differently:

	Channel
Visual	Attribute
Graphical	Dimension
Perceptual	Variable
Retinal	Cue
	Feature
	Carrier

Marks and Channels

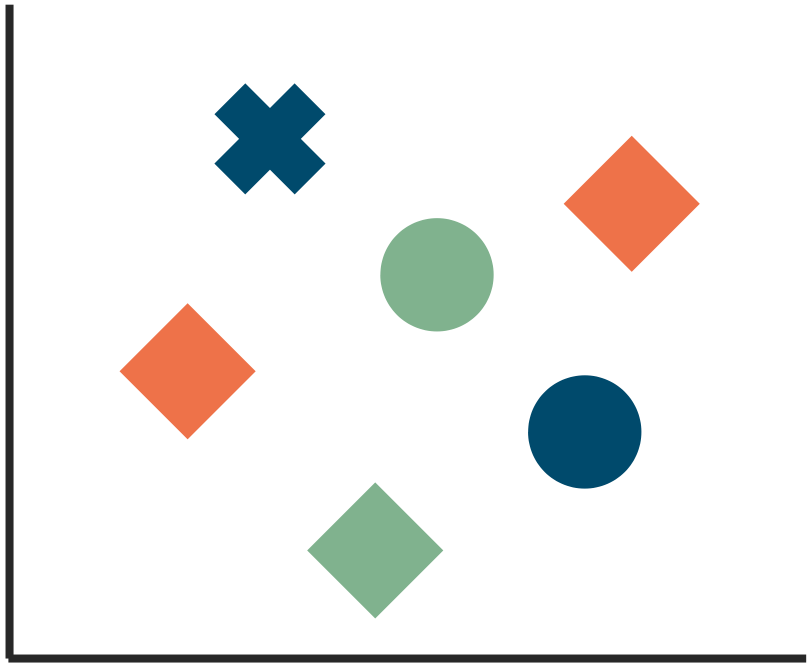
- Together, marks and channels constitute the basic building blocks of every data visualisations:

A collection of marks, each representing a *data item/link*, visually encoded by channels to reflect the value(s) of their *data attribute(s)*

Example

Marks:

Channels:



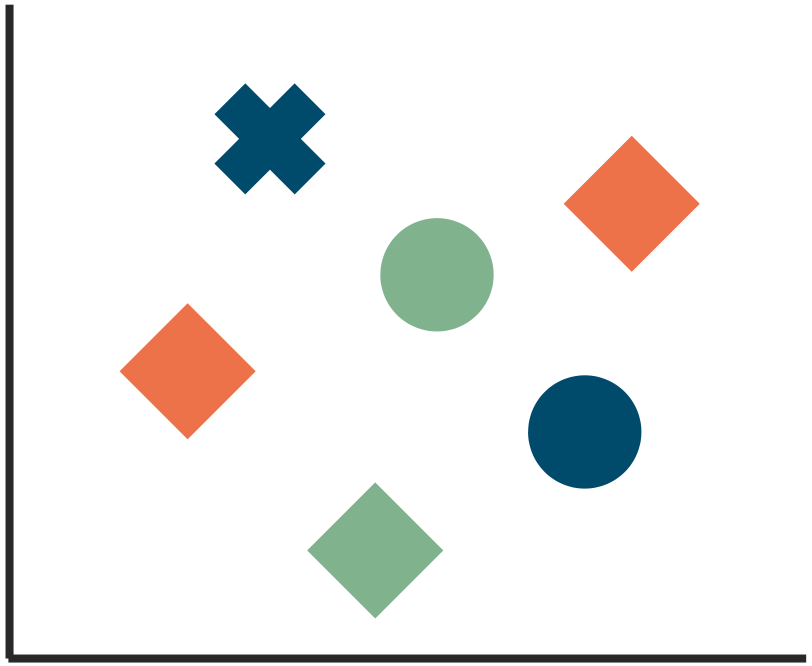
Example

Marks:

- Points

Channels:

- Vertical position
- Horizontal position
- Shape
- Colour



Types of Channels

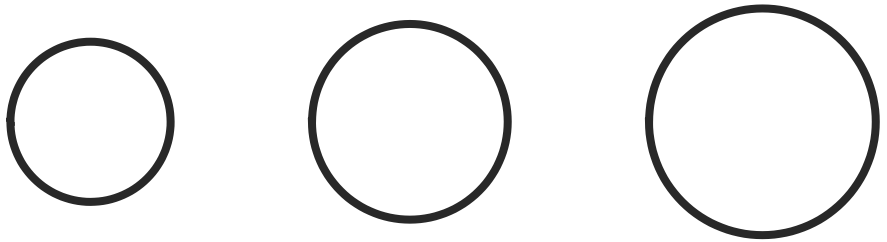
- We can distinguish two types of channel:
  - Magnitude channels for *ordered* attributes
  - Identity channels for *unordered* attributes

Magnitude Channels – Position

- The *precise location* of an item on a axis
- Can be vertical, horizontal or even in depth (3D)
- Can be aligned or unaligned

Magnitude Channels – Size

- Closely related to position, one could see size as a secondary position
- Can be length (1D), area or weight (2D), or volume (3D)
- Typically used to compare values between items

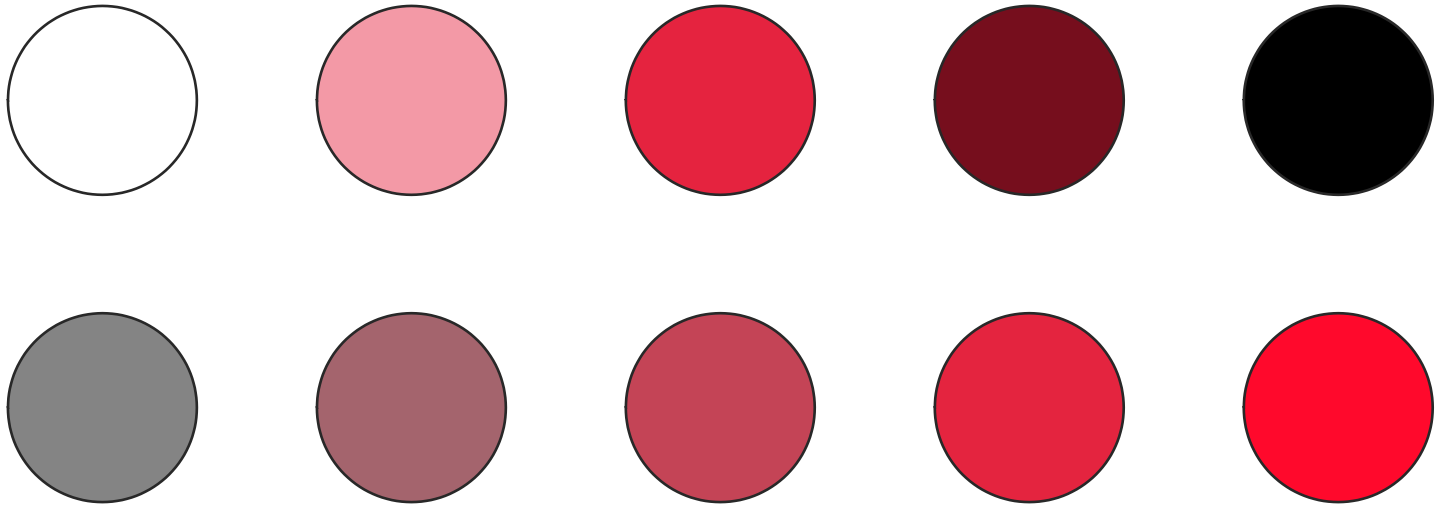


Magnitude Channels – Angle

- A.k.a. *tilt* or *orientation*
- People get less good at measuring differences between angles (but great at noticing it)

Magnitude Channels – Colour

- In particular, two aspect of colour:
  - The *luminance* (or value) – light intensity
  - The *saturation* – colour intensity



Magnitude Channels – Curvature

- Curvature mostly applies to line marks representing links
- It can be seen as an extension of angle

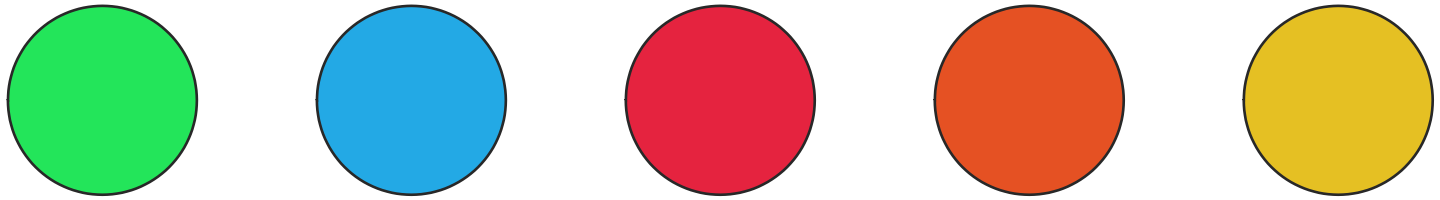
Identity Channels – Region

- Rather than position, we would refer to *region* when encoding discrete data

- The *general location* on one or more axes

Identity Channels – Colour

- This time, one aspect of colour:
  - The *hue* – the actual colour

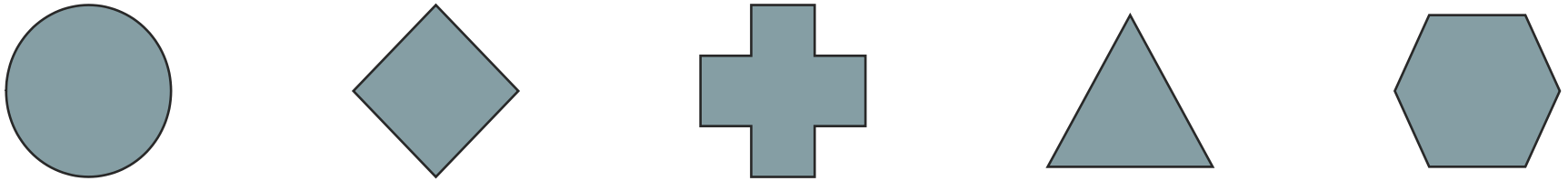


Side Note

- Colour is difficult to use effectively and raises accessibility issues
- Generally speaking, try to avoid relying on colour hue alone
- In doubt, use pre-existing scale based on luminance
  - colorbrewer
  - viridis

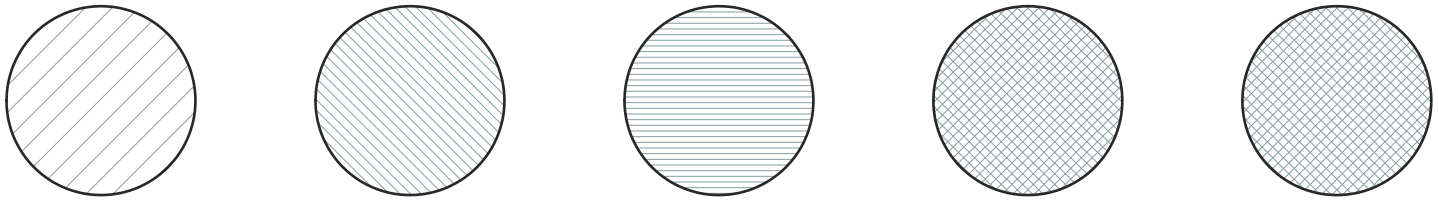
Identity Channels – Shape

- Shape can be applied to point marks (0D) mostly
- It doesn’t combine well with size



Identity Channels – Texture

- Texture is not used much nowadays
- It sometimes introduces a lot of visual noise



Channel Effectiveness

- The *importance* of a data attribute should match the channel *salience*

1. Aligned spatial position
2. Unaligned spatial position
3. Length
4. Angle
5. Area
6. Depth
7. Luminance / Saturation
8. Curvature
9. Volume

1. Region
2. Hue
3. Shape
4. Texture

Channel Effectiveness

- The *importance* of a data attribute should match the channel *salience*

1. **Aligned spatial position**
2. **Unaligned spatial position**
3. *Length*
4. *Angle*
5. *Area*
6. *Depth*
7. *Luminance / Saturation*
8. *Curvature*
9. *Volume*

1. **Region**
2. *Hue*
3. *Shape*
4. *Texture*

Attributes encoded using position will dominate the user's *mental model*

Channel Effectiveness

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There several ways to assess the effectiveness of visual channels:

- **Accuracy**: how close is *perceptual judgement* to *objective measurement*?
- **Discriminability**: are differences between items perceptible?
- **Separability**: how *independent channels* are from one another?
- **Pre-attentive processing**: how quick are distinct elements detected?

Perception

Perception

- Perception is a rich and complex area spanning several disciplines:
  - Biology
  - Neuro-science
  - Psychology

- We will only briefly explore two key concepts to data visualisation:
  - Grouping
  - Highlighting

Gestalt Principles

Gestalt Principles

- From Wikipedia:

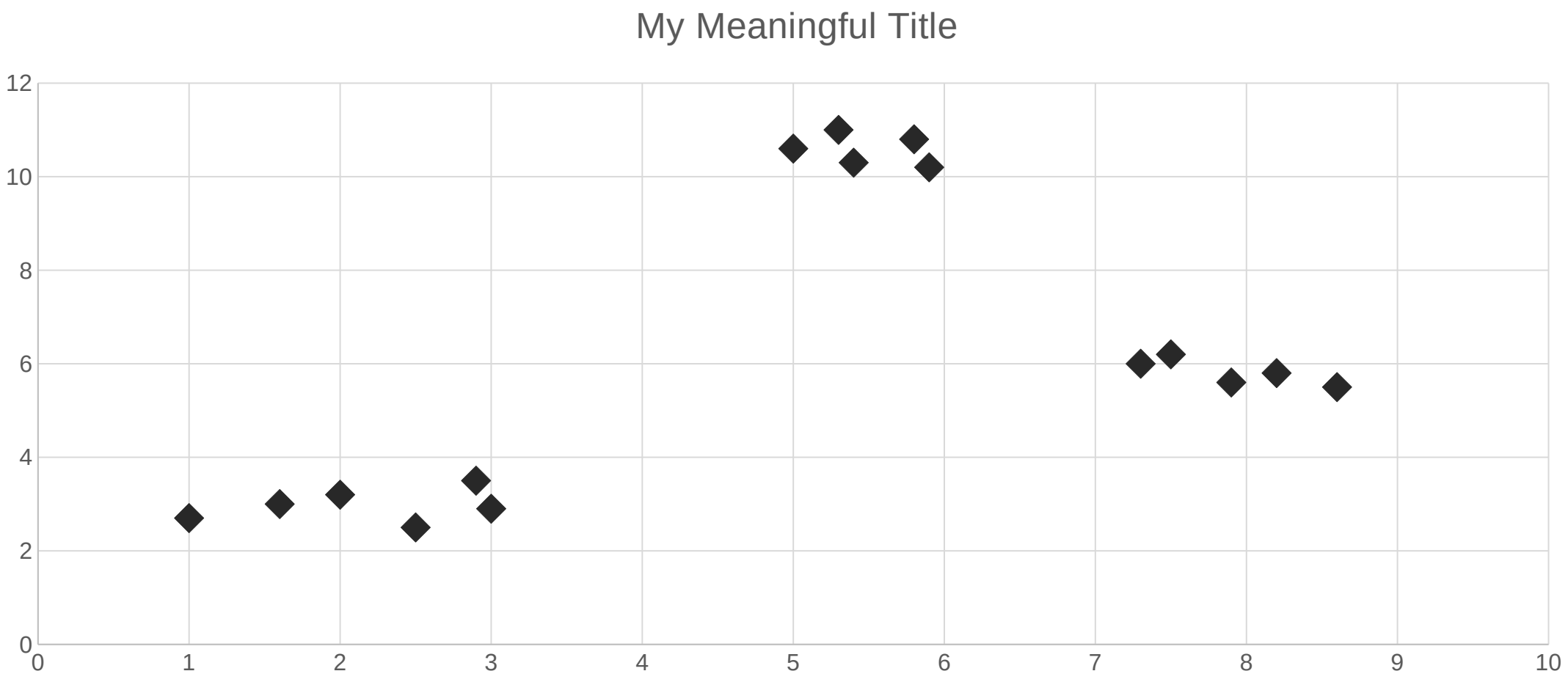
The principles of grouping (or Gestalt laws of grouping) are a set of principles in psychology to account for the observation that *humans naturally perceives objects as organised patterns*.

- Put simply:

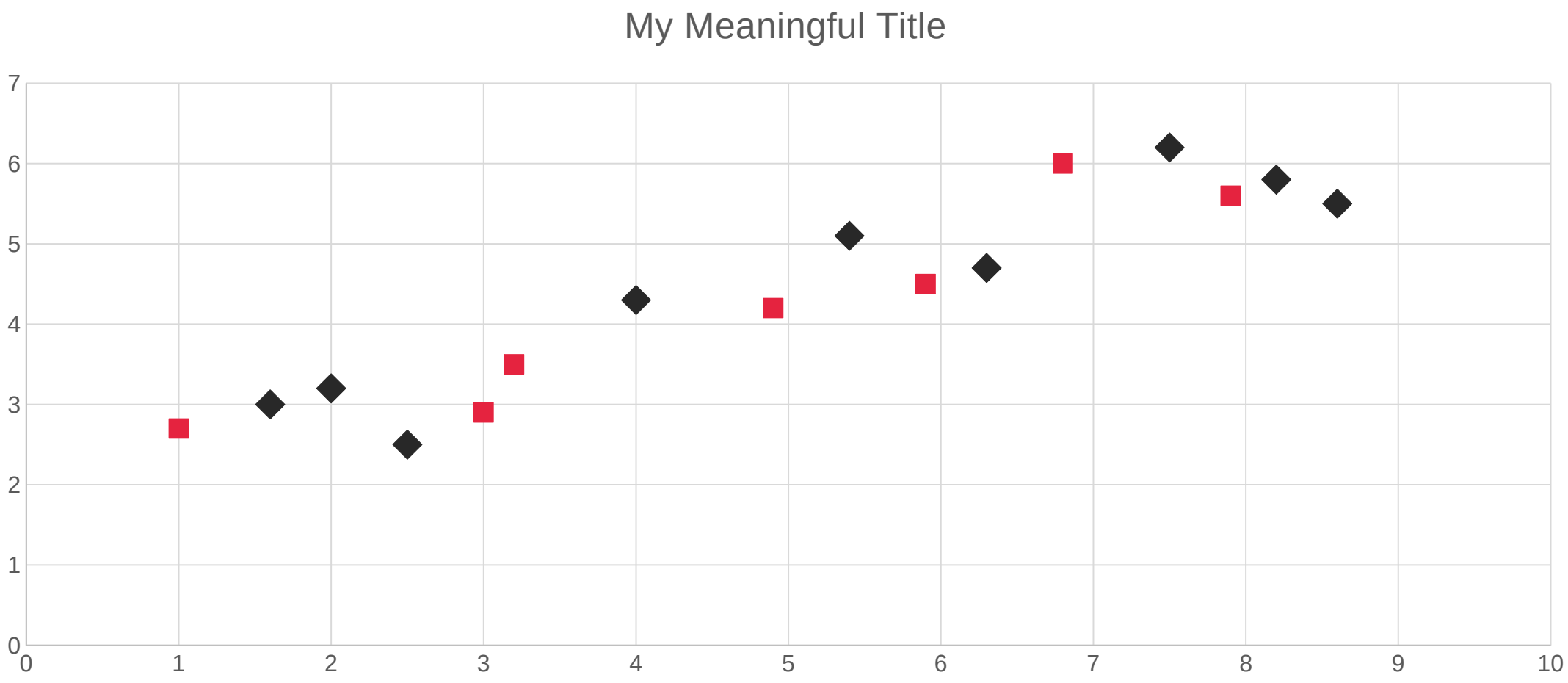
Humans want to see patterns (pareidolia)

Humans will see groups

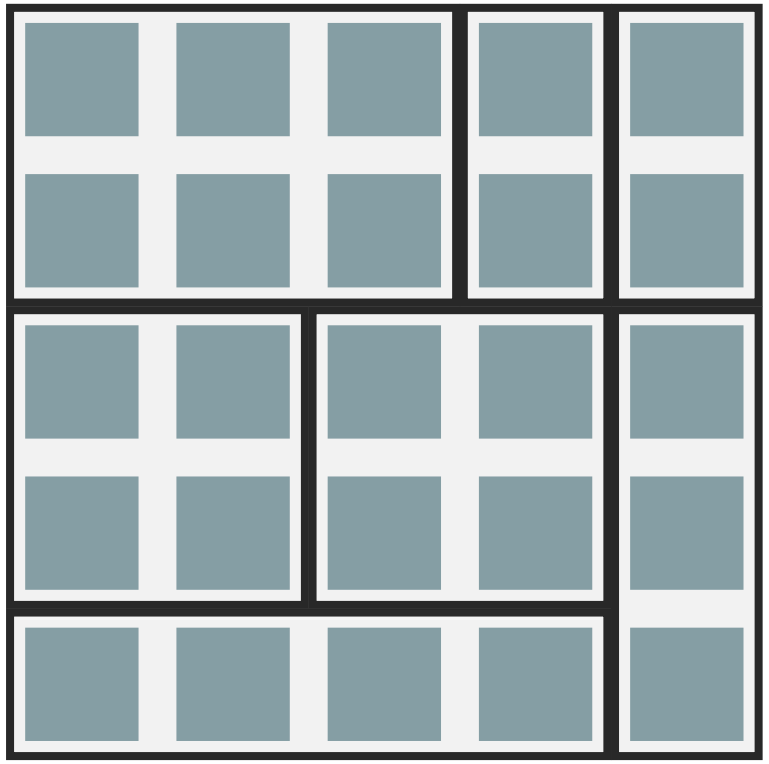
Principle of Proximity



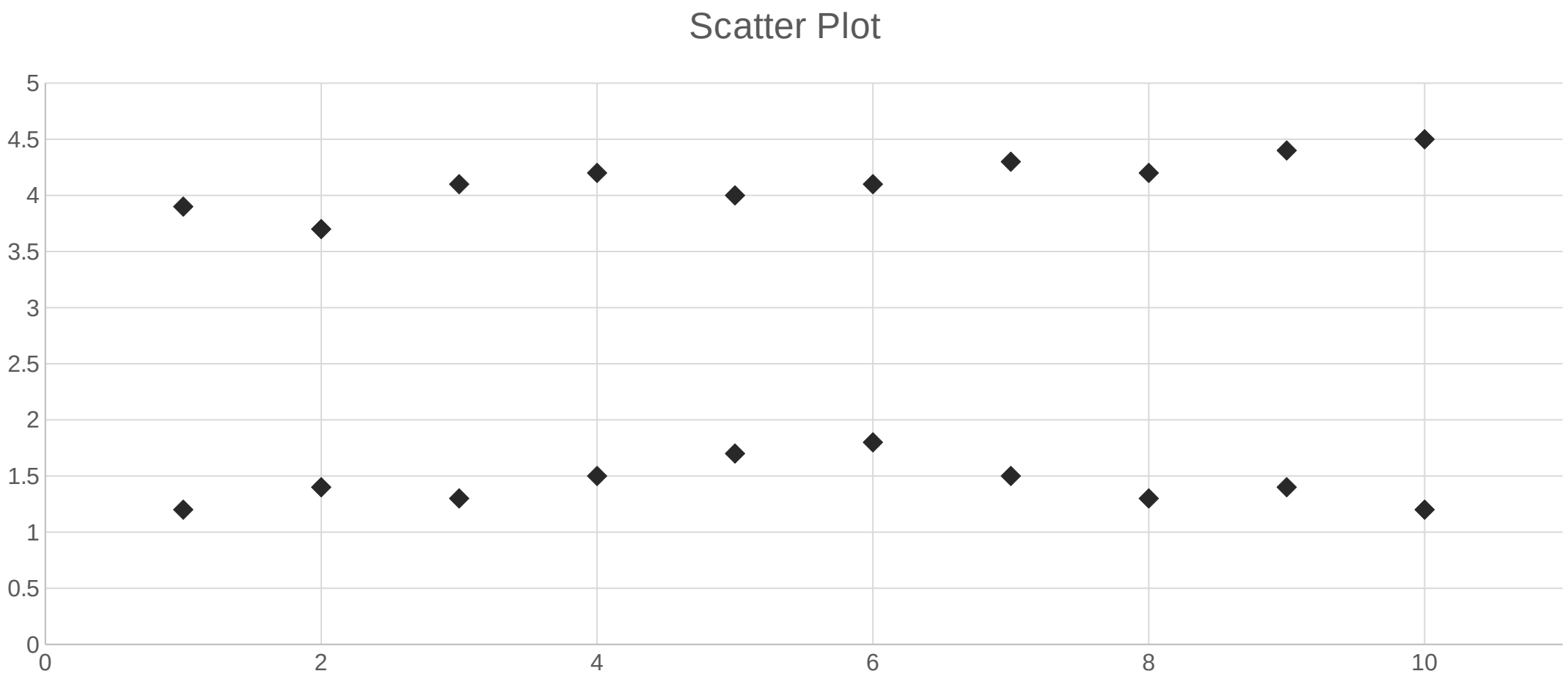
Principle of Similarity



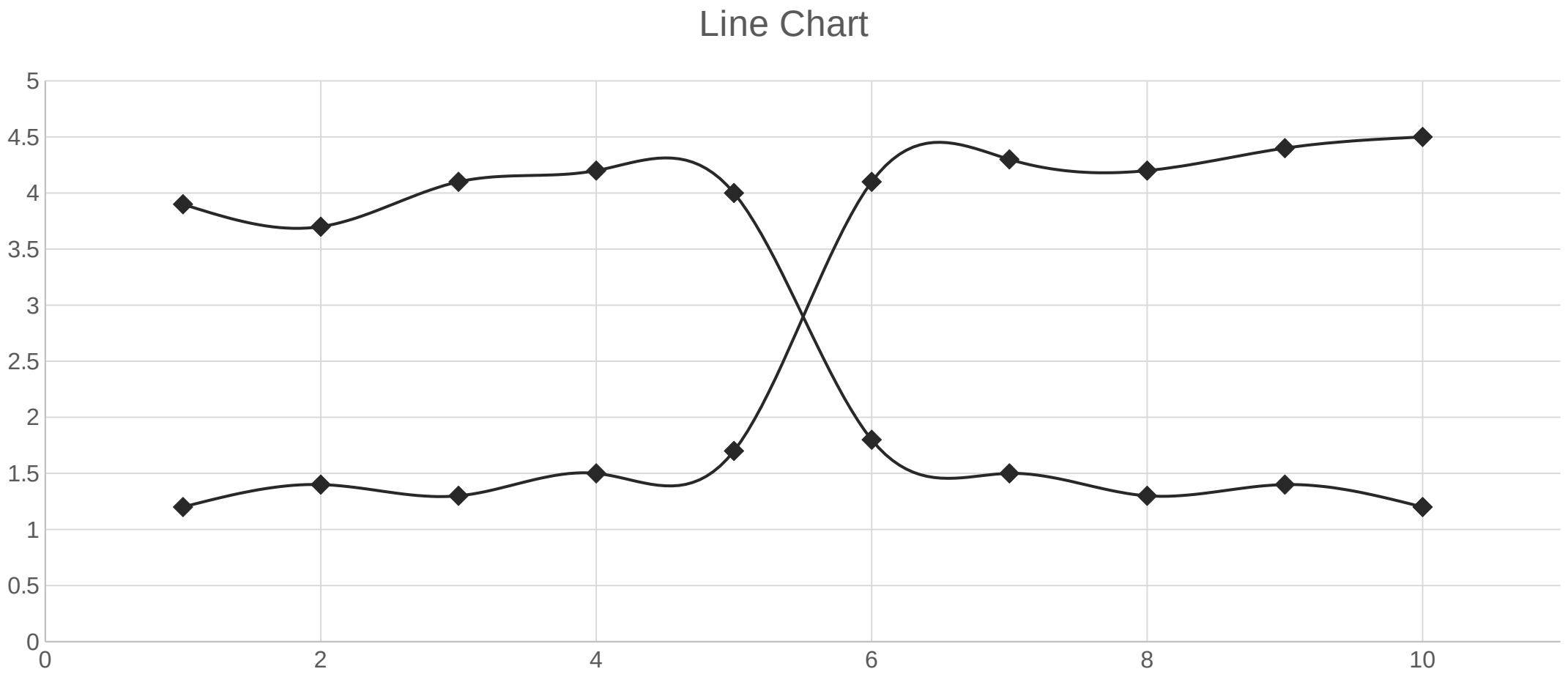
Principle of Common Region



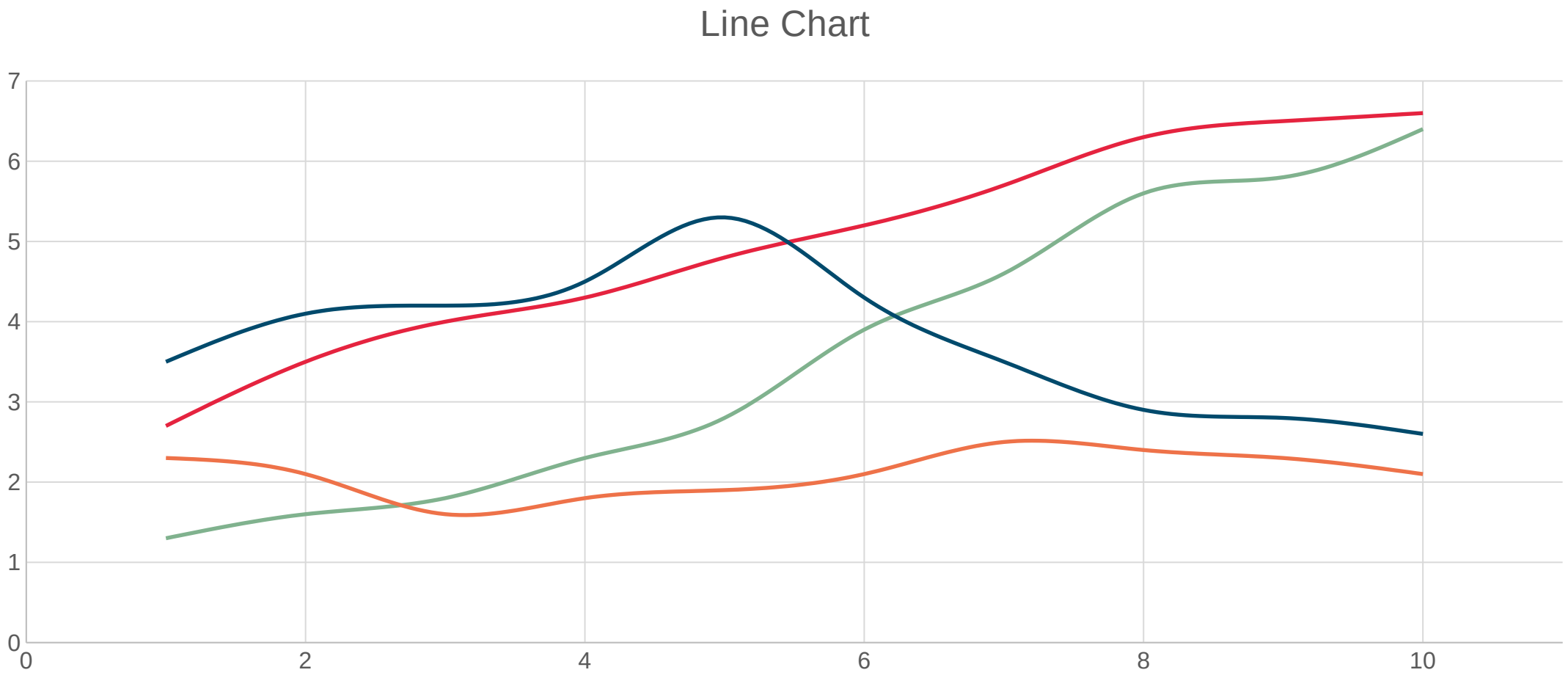
Principle of Continuity



Principle of Connectedness



Principle of Common Fate



Pre-Attentive Features

Pre-Attentive Features

- The human brain processes what it sees before it interprets it
  - And that can take some time (relatively speaking)

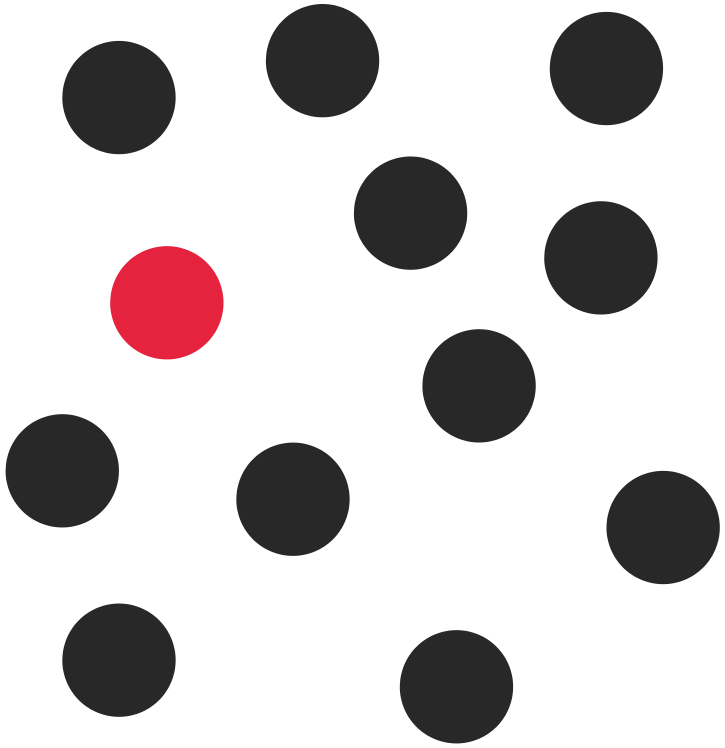
- But, some visual (and other sensory) features will be noticed before processing takes place
  - *Before* the brain pays *attention* to them

- Put simply:

Humans can’t help but notice things

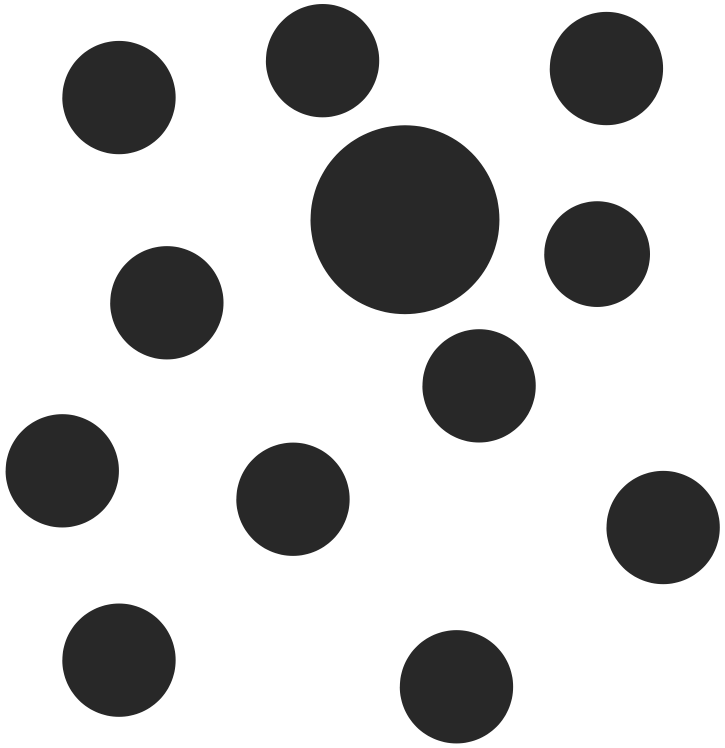
Colour

- Often used to highlight outliers or draw attention to certain data points
- The bigger the difference in luminance (value), the more obvious it becomes



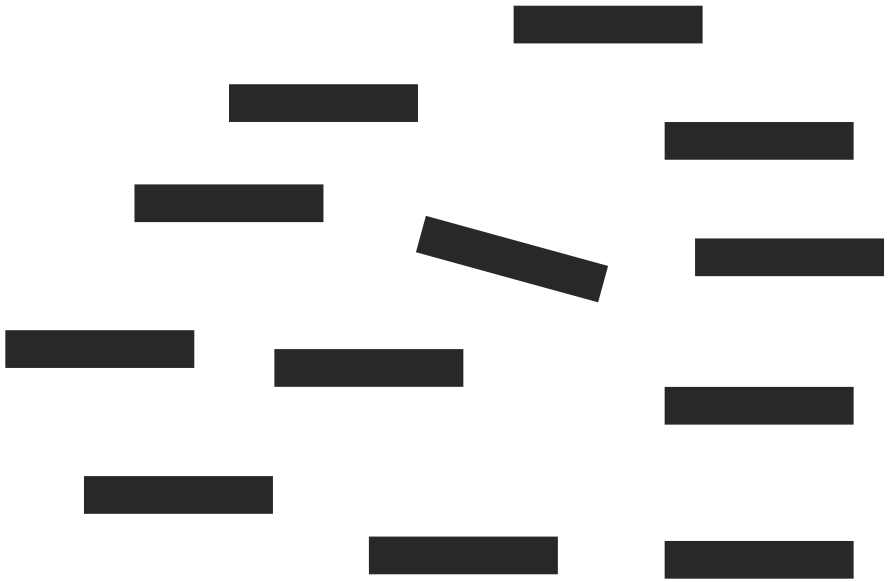
Size

- Can be used to draw attention too
- But fits better as a channel meant for comparing, measuring difference
- Again, the bigger the difference in size, the more obvious it becomes



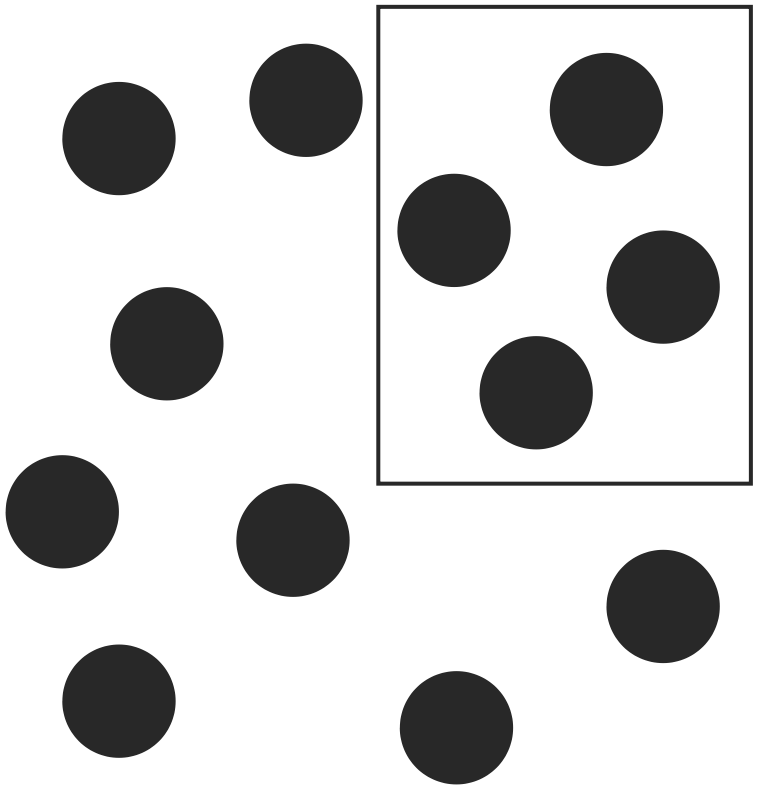
Angle

- We tend to better detect slight variations, compared to size
- But we are bad a quantifying these differences



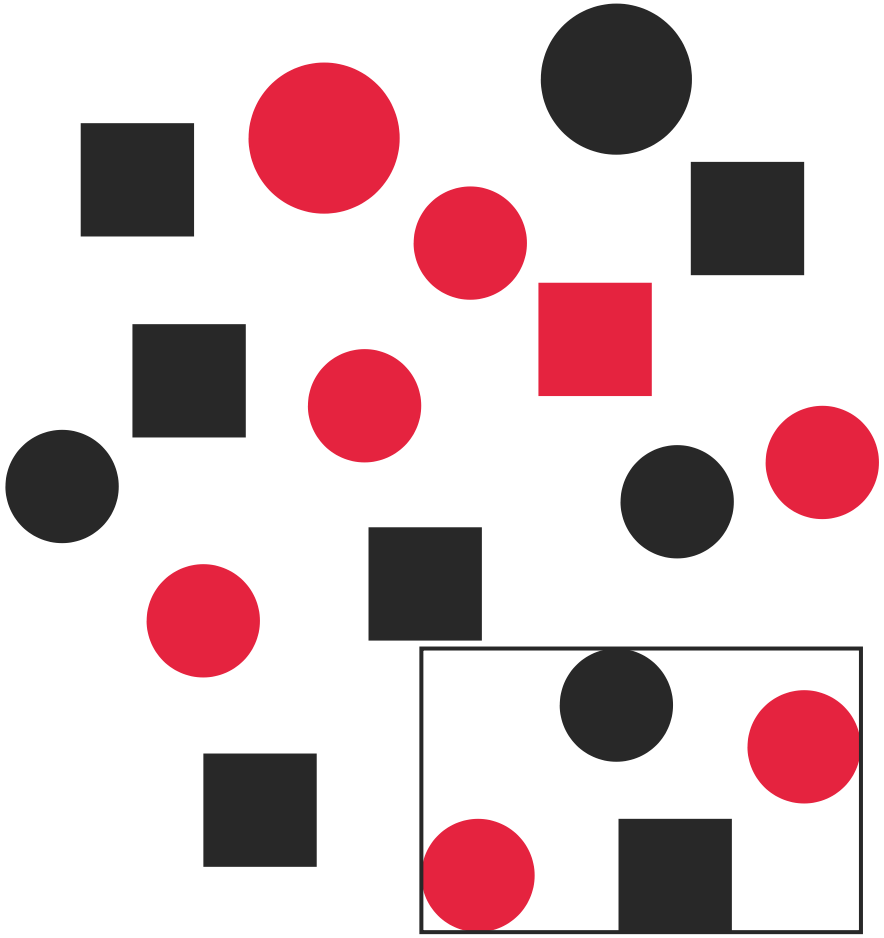
Enclosure

- Often used to draw attention to a *group* of items (Principle of Common Region)



Less is More

- When highlighting things, beware not to overdo it and have confounding features



Coordinate Systems & Scales

Coordinate Systems

- **Position** is the most important channel
  - Both for ordered (magnitude) and categorical (identity) data attributes
  - **Size** follows soon after
- It dominates the *mental model* of users
- A coordinate system will determine how we encode position and size

Coordinate Systems

- While there are many possible coordinate systems, three cover most of the data visualisation landscape:

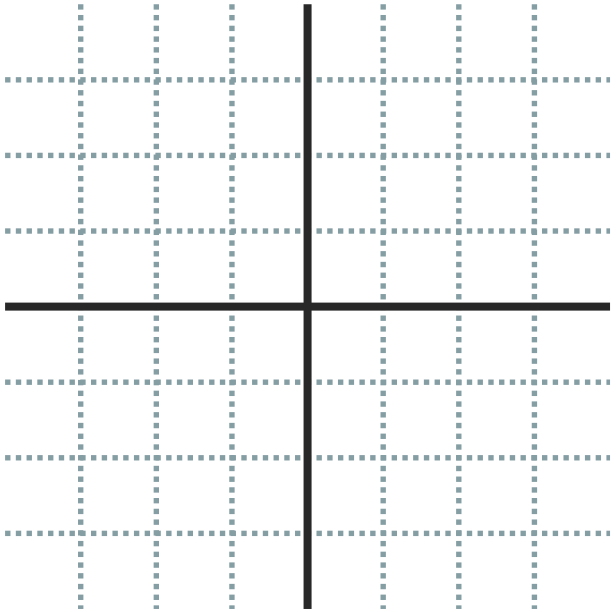
  Cartesian

  Polar

  Geographic

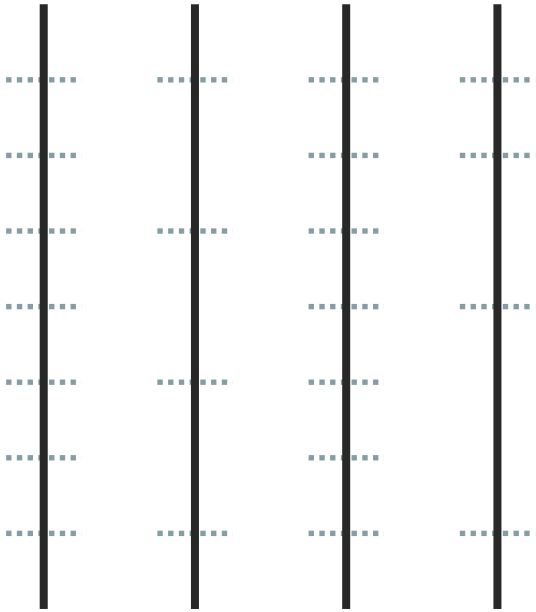
Coordinate Systems – Cartesian

- A Cartesian coordinate system is typically characterised by one or more orthogonal (perpendicular) axes
- The value on each axis determines the position of items
- The most commonly used system



Coordinate Systems – Cartesian

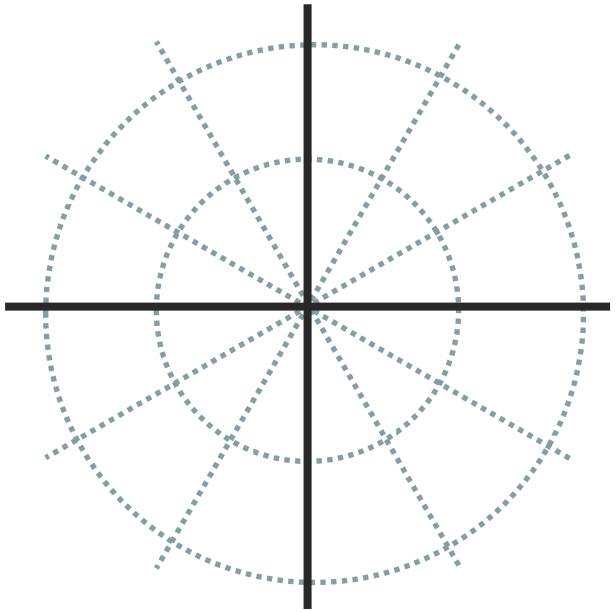
- One variation consists in having parallel axes instead
  - Which can be used to encode multiple dimensions



Coordinate Systems – Polar

- A Polar coordinate system needs two values to place items:
  - A radius – the distance from a centre point
  - An angle – the clockwise arc distance from the origin line

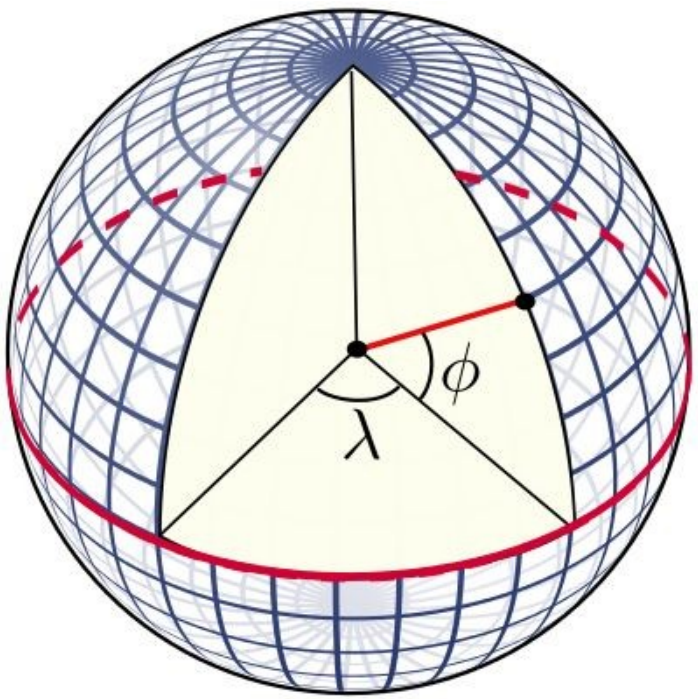
- Typically used to represent *cycles* or *wholes*



Coordinate Systems – Geographical

- As the name suggests, typical to spatial data

- Uses two angles to position a point on a sphere:
  - The latitude: angle between the point, the centre of the earth and the equator
  - The longitude: angle between the point, the centre of the earth and the Greenwich meridian

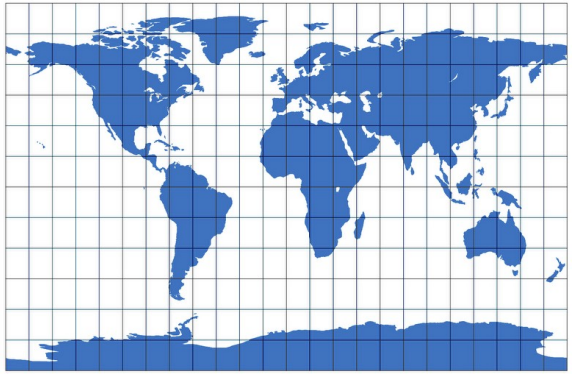


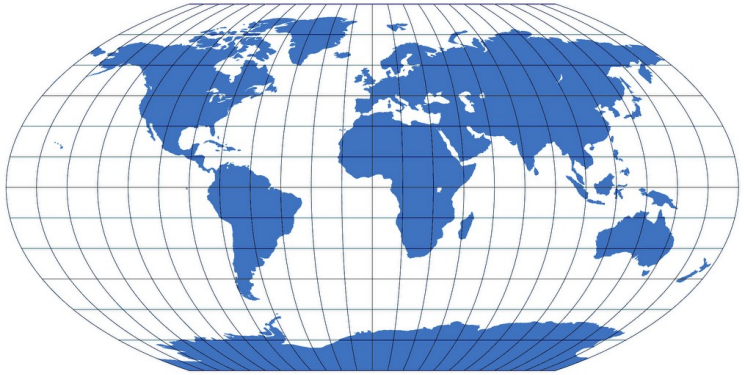
Coordinate Systems – Geographical

- The difficulty resides in showing a spherical volume, onto a two-dimensional support

  (c.f. Gauss's Theorema Egregium)

- Inevitably, distortions are introduced





Scales

- Scales are the second parameter determining the position and size of marks
- In fact, scales can be used to define any mapping from data attributes to visual channel value

Scales

- Scales can be numerical, or continuous
  - But have varying degrees of difference between intervals
  - E.g., linear, logarithmic, exponential …
- Scales can be categorical, or discrete
- Scales can be temporal

Conclusion

Key Points

- Data Visualisation are made from several components

- **Marks** are the visual units encoding data items
- **Channels** are the visual features of marks encoding data attributes

- Using visual channels introduces perceptual biases:
  - **Gestalt principles** inform on how human group items
  - **Pre-attentive features** will highlight items

- A **coordinate system** will define how items are positioned
- A **scale** will provide a mapping data values to the visual values